

However, ARM still wins by powering the most popular devices.

Vishal Dhupar, managing director-Asia South, NVIDIA, explains: "Although x86-based SoCs have been recently introduced in the mobile market with the promise of higher performance and lower power, most popular and widely used mobile devices such as iPad, HTC One X+, Microsoft Surface, ASUS Transformer Prime and Samsung Galaxy S3 are using ARM-based SoCs. This indicates that ARM CPU architecture has both the performance and power efficiency to meet the demands of mobile applications."

System-level optimisation

Since RISC architecture is simpler, developers are able to squeeze out all the performance of a RISC-based processor. Moreover, it uses far fewer transistors, which allows it to remain cool under stressful loads, thus providing consistent performance. This is one

of the reasons why Apple's software combined with their own hardware is able to perform very smoothly. Developers know exactly what they are getting into when they develop for this ecosystem and are thus able to use every iota of power in the device for their application.

Senior engineers agree that system-level optimisation wins—no matter how much money you spend on improving the implementation. Software engineers learn this early in their careers, when they understand that algorithms with a lower order of complexity often perform better than algorithms with a higher order of complexity—especially on more complex or larger problems.

"A poorly implemented algorithm that has a complexity of $O(n \log n)$ will outperform the best implemented algorithm with a complexity of $O(n^2)$ on large problems. Unfortunately, this simple logic is hard to understand for some of the largest

hardware companies. There may still be time for a company as strong as Intel to strike back," explains an experienced industry veteran from this field.

As observed in Fig. 4, when it comes to performance, the Intel Atom seems to be equally balanced among different scores. It is the floating-point performance of Samsung and Qualcomm chips that boosts them to the top of the chart. What is questionable, however, is whether it is the floating-point or memory and stream scores that could point to better real-life performance when these SoCs are used in consumer devices.

"The x86 core is better in some areas like mostly single-threaded tasks. For example, JavaScript execution and website rendering would stand out. However, in applications like audio and video playback, ARM chips have dedicated hardware codecs, so the difference is negligible," explains T. Anand, MD, Knewron.

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